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## Entity Manipulation Language - EML | RAP Series

Yogesh__Vijay
Product and Topic Expert

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2020 Nov 26 3:23 PM

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## What is EML?

Entity Manipulation Language (EML) is a part of the **ABAP language.** It means that there are new syntaxes available in ABAP to

- **control the business object's behavior** in the context of **ABAP RESTful programming model**, using ABAP code user can manipulate Business Object created using ABAP RAP model

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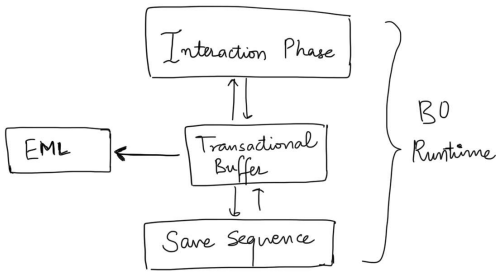
- provides a **type-save read** and **modifying access to data** in transactional development scenarios.




*Where EML comes into the picture*

## How EML syntaxes look like?

EML contains syntaxes to READ, MODIFY (Create, Update, Delete, Execute Actions) and COMMIT. Let's understand each syntax step by step

**READ ENTITIES**

P patrick_schad 46
K KevinYin 35
Y Yonggang_Yu 35
felixbartler 25
FlorianVogt 25
jeet_kapase 24
ThomasReiss 23
BiserSimeonov 21
OlgaDolinskaja 20
Burcu_Karlidag 19
View all >

defined entities in Behavior

read access

RAP BO (CDS view + Behavior)

```
READ ENTITIES OF ZTKS_I_TRAVEL_M
  ENTITY Travel
    ALL FIELDS WITH VALUE #( ( TravelUUID = '<your UUID>' ) )
  RESULT DATA(travels4)
  FAILED DATA(failed)
  REPORTED DATA(reported).
```

selection parameter

implicit params

if any key failed to get data

table contains all data

if messages are raised on BO implementation

## MODIFY ENTITIES - CREATE

modify access

BO

```
MODIFY ENTITIES OF ZTKS_I_TRAVEL_M
  ENTITY Travel
    CREATE
      SET FIELDS WITH VALUE
        #( ( %cid        = 'MyContentID_1'
             AgencyID    = '70012'
             CustomerID  = '14'
             BeginDate   = cl_abap_context_info=>get_system_date( )
             EndDate     = cl_abap_context_info=>get_system_date( ) + 10
             Description = 'Follow Development Learning' ) )

MAPPED DATA(mapped)
FAILED DATA(failed)
REPORTED DATA(reported).
```

temp ID to distinguish record

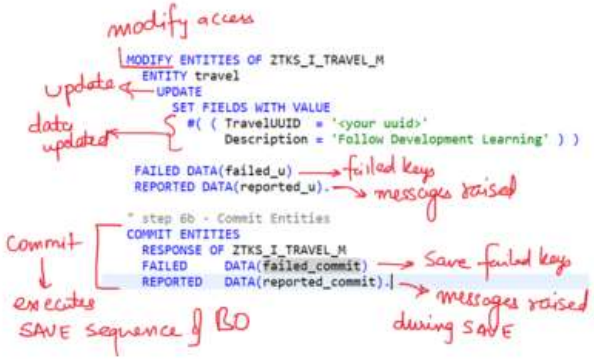
Input data

mapping b/w %cid & actual ID created

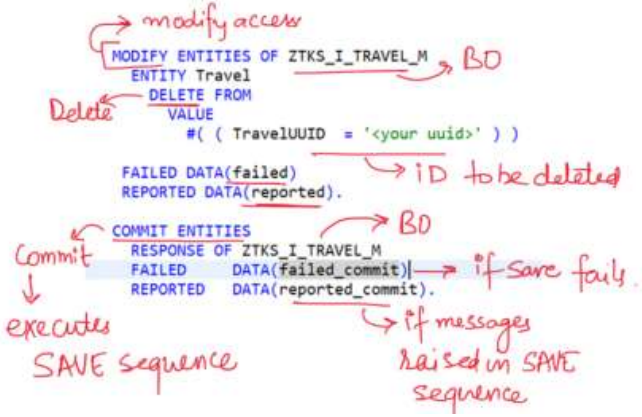
failed key → %cid keys

any messages raised.

## MODIFY ENTITIES - UPDATE




## MODIFY ENTITIES - DELETE




## MODIFY ENTITIES - ACTION

Commit Entities syntax is clubbed with Modify syntaxes to logically complete the flow. LUW concept is there in all the MODIFY syntaxes.

MODIFY syntax manipulates Transactional Buffer and READ also get data from Transactional Buffer. In the case of Managed Behavior, managed runtime automatically populates data in Transactional buffer and in case of Unmanaged Behavior, a user needs to implement Transactional READ method of Behavior Implementation

## Where is EML used?

Usually, Business objects that are implemented with the ABAP RESTful architecture based on the behavior definition and implementation of the **interaction phase** and **save sequence** in behavior pools can be consumed by means of OData protocol (Fiori UIs, or Web APIs)

Now to access them through ABAP code **EML syntax** is used.

The **standard API** is used whenever the "target" business object is statically specified. It provides code completion and static code checks. This typed API provides statements for read-only access to data (READ ENTITIES), as well as for modifying data access (MODIFY ENTITIES) and for triggering the save sequence (COMMIT ENTITIES), these all we saw above

The **generic API** is typically used for generic integration of business objects into other frameworks, such as the Cloud Data Migration Cockpit or the Process Test Framework.

## What is the Use Case?

One of the uses cases of EML is the writing of test modules as ABAP Unit tests. As ABAP application developer, it gives you the ability to test the transactional behavior of business objects for each relevant operation that is defined in the behavior definition.

Another use-case is cross BO integration. Executing CRUDA operation of RAP based BO in another RAP based BO' Implementation in exits like determination, validation, actions, etc.

## From which release it is available?

EML - Entity Manipulation Language is available after **1902 Cloud** release and **1909 on-premise** release

## Few points to note

1. Multiple modify elements can be used in one MODIFY statement. The sequence of the operations is irrelevant.

2. EML syntaxes for LUW concepts

3. Authorizations will be automatically handled by these syntaxes, if a user needs to escape authorizations then just add syntax **- IN LOCAL MODE**

4. READ by associations also works. From a parent entity, child entity can be read and vice versa.

Kindly share your thoughts on this blog and I will continue writing blogs on RAP topic.

Regards,

Yogesh Vijay

RAP Trainer, Development Learning

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## 21 Comments

S **sergey_muratov**
Participant

2020 Nov 27 4:43 AM

2 Likes

Hi
My question is a little out of the question. It seems to me that expanding the syntax of a language for each function instead of building a library on top of an existing language is a strange solution?

**Yogesh__Vijay**
Product and Topic Expert

2020 Nov 27 5:11 AM

1 Like

Hi Sergey,

In some sense you are right, however, when there is a new Programming Model came into the picture – which includes new artefacts like we have Behavior Definition, Behavior Implementation, etc then adding new syntaxes specific to that model is I guess fine.

I can think of that there must be some limitation to build on top of existing language one. Anyways I will try to pass on this question to the relevant team and hope we get some answers to it.

Thanks,

Yogesh Vijay

**s7oev1**
Active Participant

2020 Nov 27 10:54 AM

0 Likes

I very much agree with this point. Why not take the modern approach used in, e.g., Java, Python and JS, where the core functionality is extended through libraries rather than adding new syntax to the language?

**JaimeRodriguez**
Participant

2020 Dec 03 5:00 PM

0 Likes

I tried to build a simple development, just to update a table depending on some vendor conditions, so I included the management in a BAdI.

I had many problems at the moment of saving the data:

- I cannot manage the rise of the COMMIT, so I can't do COMMIT ENTITIES
- Another problem, the vendor data is updated from CIG via API, this calls all saving process during the COMMIT, so I can't perform the saving of data

Workaround to save all data was implementing a new local class with data to save and a method to handle COMMIT, and from behaviour saver local class I call to this one.

```
CLASS lcl_saver DEFINITION.

  PUBLIC SECTION.
    CLASS-METHODS get_instance
      RETURNING
        VALUE(ro_result) TYPE REF TO lcl_saver.

    METHODS add_data
      IMPORTING
        it_create TYPE arbcig_an_vendor_t OPTIONAL
        it_update TYPE arbcig_an_vendor_t OPTIONAL
        it_delete TYPE arbcig_an_vendor_t OPTIONAL.

    METHODS save.

    METHODS cleanup.

  PRIVATE SECTION.
    CLASS-DATA:
      go_saver TYPE REF TO lcl_saver.

    DATA:
      mt_create TYPE arbcig_an_vendor_t,
      mt_update TYPE arbcig_an_vendor_t,
      mt_delete TYPE arbcig_an_vendor_t.

    METHODS handler_transaction_finished
      FOR EVENT transaction_finished OF cl_system_transaction_state
      IMPORTING
        kind.

ENDCLASS.
```

```
CLASS lsc_Vendor DEFINITION INHERITING FROM cl_abap_behavior_saver.
  PROTECTED SECTION.
    METHODS save REDEFINITION.
    METHODS cleanup REDEFINITION.

ENDCLASS.

CLASS lsc_Vendor IMPLEMENTATION.

  METHOD save.
    lcl_saver=>get_instance( )->save( ).
  ENDMETHOD.

  METHOD cleanup.
    lcl_saver=>get_instance( )->cleanup( ).
  ENDMETHOD.

ENDCLASS.
```

This way, I can save the data at COMMIT and COMMIT ENTITIES.

The solution to write the changes if EML is called during COMMIT, is to perform DB modifications directly in the CREATE/UPDATE/DELETE methods when sy-oncom = 'P'

I think that EML needs to take a look at this.

Yogesh__Vijay
Product and Topic Expert

2020 Dec 03 5:05 PM

0 Likes

Hi Stoyko,

What you are talking about is something totally different. Let say in python or java, if they introduce another library then you have new syntaxes right? For example, at one point of time if string library is added then string functions like concat, etc came into the picture. Similarly, with the new programming model, we have new artefact as Behavior definition and now to manipulate BO based on RAP there is new syntax. There is no existing library which was there which can be enhanced.

What's your thought?

Yogesh__Vijay
Product and Topic Expert

2020 Dec 03 5:13 PM

0 Likes

Hi Jaime,

Doing this will break the LUW, we have actions also along with basic operations and an option of Unmanaged SAVE. This will create a problem in view

Here Commit work is implicitly handled by framework and ideally BAdI's and BAPI or any FM should not contain COMMIT WORK inside.

Any other thought?

J

**JaimeRodriguez**
Participant

2020 Dec 03 5:29 PM

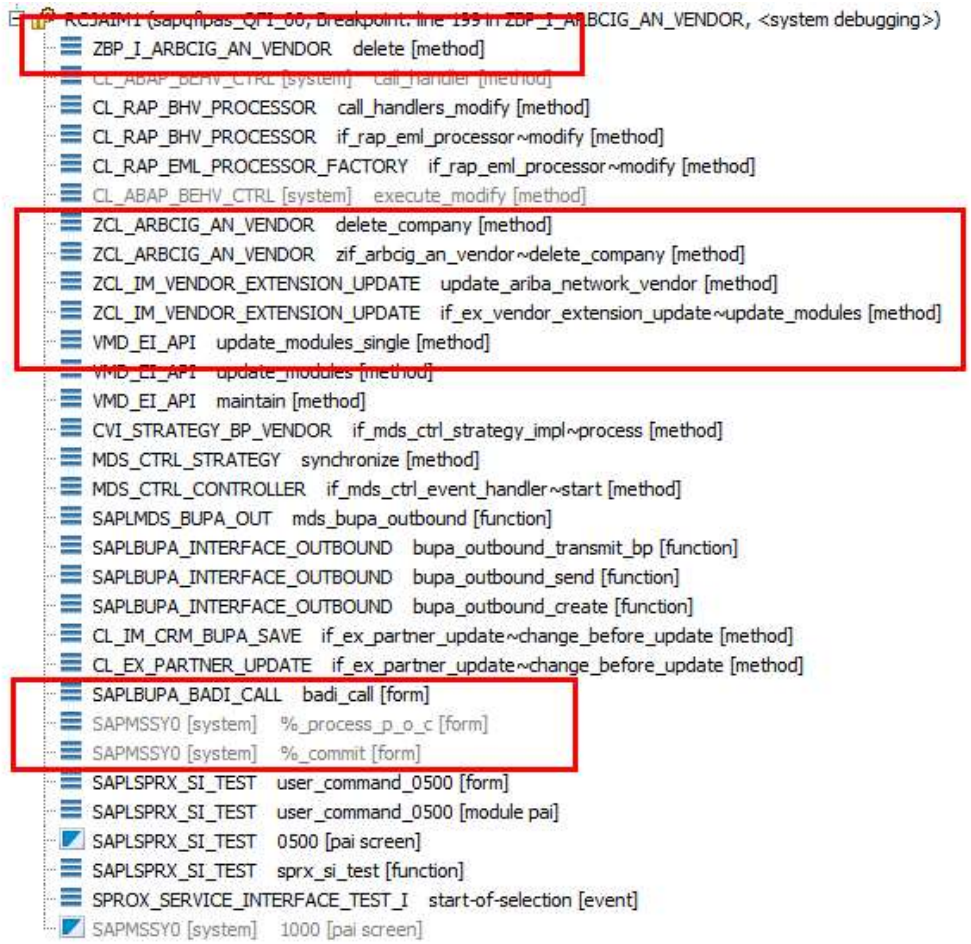
1 Like

Hi Yogesh,

I think that I didn't express myself correctly.

I am not doing the COMMIT by my self, it is performed by a proxy class.

This is the call stack:




The "delete_company" method:

```
METHOD delete_company.

  MODIFY ENTITIES OF zi_arbcig_an_vendor
    ENTITY Vendor
      DELETE FROM VALUE #( ( Company = iv_Company
                             Vendor  = mv_vendor ) ).

ENDMETHOD.
```

I can't execute the COMMIT ENTITIES in this situation

s7oev1
Active Participant

2020 Dec 03 8:06 PM

0 Likes

Hey, Yogesh,

I agree my choice of words could be clearer – I meant new **core** syntax. Other than this, I think my point is pretty clear – as you mention, Python or Java introduce new features using libraries. The **core** syntax of the language is the same, however. And very simple at that. There's mainly just variables, arithmetic operations, conditionals, loops, classes, and functions/methods.

Instead, what we see here in ABAP RAP is very different. Statements like READ ENTITIES OF, FAILED, REPORTED, etc. are now part of the **core** syntax. *The language itself has been modified*. If they were instead provided through a library as they would in Java or Python, it would be something like this:

rap.read_entities_of(), rap.failed(), rap.reported()

etc.

See the difference?

Another example would be how sin() in ABAP is a function part of the **core** syntax, but in Java it's provided through the Math library. It's not part of Java's **core** syntax.

Best,

Stoyko

s7oev1
Active Participant

2020 Dec 03 8:28 PM

1 Like

Also, if you're interested check out this video, from 9:15:

https://www.coursera.org/lecture/nand2tetris2/unit-3-11-perspective-29aEp

It speaks exactly about the topic at hand. More specifically, it mentions how Java is extended through libraries but the language itself has been practically unchanged since 1996 🙂

Yogesh__Vijay
Product and Topic Expert

2020 Dec 04 4:33 AM

0 Likes

That's a very good explanation, you are right what we have in JAVA or python.

Here we can have libraries (usually we have using standard classes) but adding another artefact which simplifies user experience - If you see behavior definition, it is easy to understand what all operations are available for my Business Object.

otherwise behind the scene things are same.

former_member186021
Participant

2020 Dec 13 2:04 PM

1 Like

Hi Yogesh,

That's really helpful, thanks for sharing!

Y

yonas
Advisor

2021 Apr 01 11:21 AM

0 Likes

Hi Yogesh,

Thanks for the great sharing!

We are developing a prototype of the standard product, in current desgin, we want to directly provide a class with CRUD methods which developed with EML to interact with our RAP model(Managed mode), then expose this class as a API to our consumers.

In your opinion, is this kind of desgin acceptable?

Regards

Yogesh__Vijay
Product and Topic Expert

2021 Apr 05 3:27 PM

1 Like

Yeah, this kind of design is acceptable, it's like a wrapper. So ideally these EML syntaxes themselves work as API's but if you want to simplify the terminologies then its fine.

Just remember EML syntaxes works on KEYS, I mean you can read data only based on KEY fields, not like SQL selects.

Y

**yonas**
Advisor

2021 Apr 06 7:32 AM

0 Likes

Hi Yogessh,

Yes, we want our API consumers to use it as they used to use other normal classes, so we add a wrapper to hide EML syntaxes(e.g. MODIFY ENTITIES OF xxx)

Thank you for your comments and further reminder!

Regards

**Former Member**

2021 Aug 01 7:08 PM

3 Likes

This is incomplete. you did not talk about the deep create, modify things. Also the main problem is not EML it's quite easy the problem is when using it in RAP implementation- what is %key and %control when to use what those documentations are lacking.

**former_member462342**
Discoverer

2021 Nov 25 2:58 PM

0 Likes

I will do exactly the same thing in my project. But apart from the specific API, we also would like to separate it for testing purposes. This way we can mock the BO results in the productive class, instead having to introduce data in DB and then test.

This way we will separate the tests: the EML entity will be tested in the RAP class.

The productive code will only takes care of the main logic, BO results will be mocked.

Regards, Raquel

**former_member839746**
Explorer

2023 Feb 01 11:35 AM

0 Likes

Hi,

this is quite useful blog.

I have a query..I have raised an error in precheck method of the behaviour definition, and when I try to read the entities, the error that was raised earlier was not getting captured in the reported data..

Could you please let me know, if there is any limitation in the precheck?

M **mlewis**
Explorer

2023 May 02 3:08 PM

0 Likes

HI

I have a question regarding the Cross BO Integration Use Case. How should the Commit Entities be handled? Calling Modify/Commit from within another Behaviour implementation throws an error.

Many thanks

**Yogesh__Vijay**
Product and Topic Expert

2023 May 03 9:02 AM

0 Likes

If you are using Managed BO's then it is handled out of the box, however, if another case then you might need to create a RFC where you can handle Commit separately.

M **mlewis**
Explorer

2023 May 03 9:13 AM

0 Likes

Thanks Yogesh.

I guess my issue arises because I have wrapped the logic for the BO EML in a class and calling the class from within the Behaviour Implementation. Should the logic be kept in the Behaviour implementation? Why should it matter if it is in a separate class?

G

**giammaria-pace**
Participant

2023 Jun 07 11:41 AM

0 Likes

Thanks Yogesh.helpful blog

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PREVIEW

Paragraph

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